Lesson plan: Validate Before You Estimate

Using AI to label text and link records in social science research. 45 minutes, graduate students in the social sciences and humanities.
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Learning objectives

- Explain why a classification prompt is a measurement instrument, and what changes when the codebook goes into it.
- Design a validation sample and read accuracy, kappa, precision and recall as measurement error.
- Explain why high accuracy does not guarantee an unbiased estimate or a valid comparison between groups.
- Read a bias-corrected estimate (prediction-powered inference) that combines AI labels with a small hand-coded sample, and choose how many documents to hand-code from an interval curve.
- Apply the same recipe to record linkage: rules for the easy pairs, an AI adjudicator for the ambiguous band, a reviewed sample to measure the error.

Prerequisites, materials, fit

- None for the room exercises (hand-coding, voting, reading a curve); Python Fundamentals recommended to run the notebook afterward; no AI experience required.
- Data: Comparative Agendas Project congressional bills (1,650 titles, 75 per Congress 1973 to 2016, human-coded); Febrl person records (synthetic, 5,000 true links known). All model outputs pre-computed and cached (6,600 labels, 2,966 pair decisions; model, prompt version, tokens and timestamp logged). Runs on DataHub, Binder or a laptop without an API key.
- Catalog fit: builds on Python APIs for Large Language Models (mechanics) and Prompt Engineering; sits next to Qualitative Coding with AI Tools (the passage-by-passage audit) as the measurement step; leads into a full Record Linkage workshop. Not covered: model internals, fine-tuning, inductive coding, agent pipelines. The repository follows the D-Lab template so a Data Science Fellow can teach it after one rehearsal.

Sequence

Min	Block	Instructor	Participants (exercise goal)
0-5	Hook and poll	A trend chart built from AI labels; Desai, Card and Jacobs (2026) on validation practice; the one idea	Poll by hand: used an LLM to code? checked it? did the check change a number?
5-13	Codebook to prompt	Three instruments for one task; structured output; logging; nothing in a prompt is scored	Challenge 1: code six titles by hand, compare with a neighbor (feel the ambiguity; set the human ceiling)
13-24	Validate	Accuracy and kappa for P1, P2, P3 (73%, 87%, 88%); confusion matrix; precision and recall by class; error by party and era	Challenge 2: bet on the winning prompt; then vote on three disputed titles (model wrong, human wrong, or codebook ambiguous) and say what to change
24-32	Estimate	Naive vs hand-coded vs corrected share of health bills; a 2.5-point partisan gap the humans do not see dissolves after correction; the scale analogy; the PPI line; the era chart and what 63 to 72 titles per era can and cannot say	Challenge 3 (exit ticket): read the interval curve and choose how many titles to hand-code per group to see a 3-point gap (checkable answer: about 400 per group, 850 without the model)
32-40	Link records	Same recipe: rules sort pairs, the model adjudicates the band; precision and recall of four policies; the downstream mobility rate (12.2% to 22.3% vs truth 15.6%)	Question: lose pairs or invent them? Decide for the number you want to know
40-45	Close	What we know works; what to report; privacy and cost; key points	Take-home challenge on their own data; exit tickets collected

Teaching approach and assessment

- Backward design from the decisions a student makes in a real project; predict-observe-explain before every result is shown; an exercise every ten to twelve minutes (the D-Lab guide asks for at least one every fifteen); colloquial language, each term explained when first used; it is OK not to know, said out loud.
- Formative assessment: opening poll, bell questions as attention checks, Challenge 3 as an exit ticket with a checkable number; the take-home challenge links to D-Lab consulting.
- Ethics and evaluation inside the method: differential error by group, what data may leave the machine, cost and time, reporting standards.
- Contingencies: cached outputs and a saved dry run replace any live call; the handout makes the exercises work with or without laptops. The 40-minute mock version (overview plus lesson) folds the precision-recall and reporting slides into their neighbors and takes a minute off each challenge; a 35-minute lesson also drops the four-policies chart. The era chart always stays, because it answers the opening hook.